



CSI142 Object-Oriented Programming

Arrays and ArrayList in Practice

Week 2 • Lecture 5

Iteration patterns • common errors • command-line validation

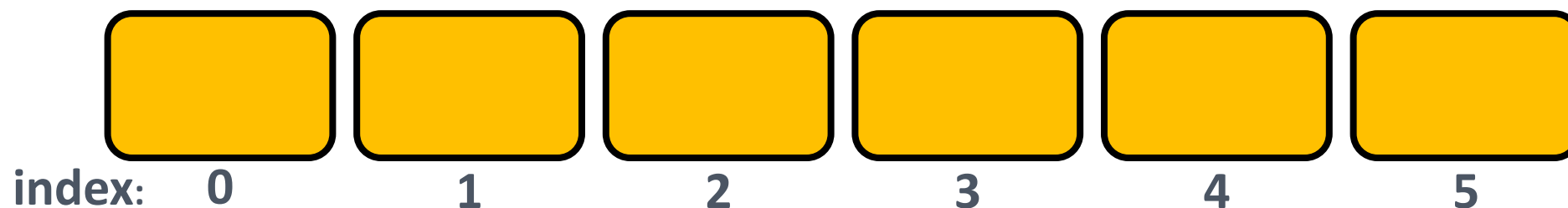
Wed 4 Feb 2026 • 08:00–09:00 • Room 252-008



- Iterate arrays safely using length and correct bounds
- Use `ArrayList<T>` operations: add, get, set, remove, size
- Choose between array and `ArrayList` for a given scenario
- Validate command-line arguments before use



- Switch fall-through (missing break)
- Infinite do-while loop (exit condition never met)
- `ArrayIndexOutOfBoundsException` (bad index)
- `args[0]` used when `args.length == 0`



Key rules:

- Valid indices: 0 to length-1
- Use `arr.length` (not a magic number)
- Off-by-one bugs are the most common array bugs



```
1 int[] scores = { 72, 81, 90, 65 };
2
3 // classic for loop
4 for (int i = 0; i < scores.length; i++) {
5     System.out.println(scores[i]);
6 }
7
8 // enhanced for (read-only)
9 for (int s : scores) {
10     System.out.println(s);
11 }
```

Key points

- Use $i < \text{length}$
- Enhanced for is simpler for reading
- Keep loop bodies small



```
1 import java.util.ArrayList;
2
3 ArrayList<Integer> scores = new ArrayList<>();
4 scores.add(72);
5 scores.add(81);
6
7 System.out.println(scores.get(0));
8 System.out.println("size=" + scores.size());
9 scores.remove(0);
```

Key points

- Remember to import
- Use <Integer> for numbers
- Indices shift after remove



```
1 ArrayList<Integer> values = new ArrayList<>();
2 while (true) {
3     int v = in.nextInt();
4     if (v < 0) break;    // sentinel
5     values.add(v);
6 }
7
8 int sum = 0;
9 for (int v : values) sum += v;
10 double avg = values.isEmpty() ? 0.0 : (double)
sum / values.size();
11 System.out.println("avg=" + avg);
```

Key points

- Sentinel ends input
- Handle empty list
- Compute using size()



Use an array when...

- Size is fixed (e.g., 12 months)
- Memory footprint matters
- Simple indexing is enough

Use ArrayList when...

- Size is unknown (user input)
- You will add/remove items
- You want list operations



```
1 public static void main(String[] args) {  
2     if (args.length == 0) {  
3         System.out.println("Usage: java App <mode>");  
4         return;  
5     }  
6  
7     String mode = args[0];  
8     System.out.println("mode=" + mode);  
9 }
```

Key points

- Check args.length
- Print a helpful usage message
- Return early on invalid input



- Find the bug: `for (int i = 0; i <= arr.length; i++)`
- Find the bug: `avg = sum / list.size()` when list is empty
- Find the bug: `args[0]` without checking `args.length`
- Exit ticket: choose array or `ArrayList` for a scenario and justify